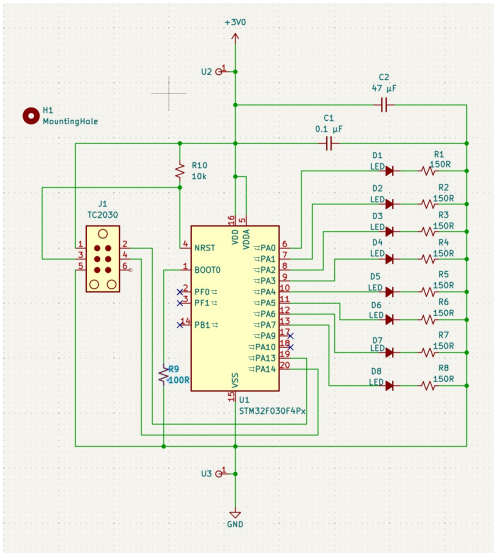
V_I



Electrical characteristics STM32F030x4/x6/x8/xC

Table 19. Current characteristics

Symbol	Ratings	Max.	Unit
ΣI_{VDD}	Total current into sum of all VDD power lines (source)(1)	120	mA
ΣI_{VSS}	Total current out of sum of all VSS ground lines (sink)(1)	-120	
$I_{VDD(PIN)}$	Maximum current into each VDD power pin (source)(1)	100	
$I_{VSS(PIN)}$	Maximum current out of each VSS ground pin (sink)(1)	-100	
$I_{IO(PIN)}$	Output current sunk by any I/O and control pin	25	
	Output current source by any I/O and control pin	-25	
$\Sigma I_{IO(PIN)}$	Total output current sunk by sum of all I/Os and control pins(2)	80	
	Total output current sourced by sum of all I/Os and control pins(2)	-80	
$I_{INJ(PIN)}$(3)	Injected current on FT and FTf pins	-5/+0(4)	
	Injected current on TC and RST pin	± 5	
	Injected current on TTa pins(5)	± 5	
$\Sigma I_{INJ(PIN)}$	Total injected current (sum of all I/O and control pins)(6)	± 25	

80 mA maximum draw
↳ 70% absolute max : 56 mA
↳ $\frac{80\ mA}{56\ mA} = 1.429$, SF

56 mA / 8 LEDs = 7 mA/LED
↳ 4.45 ed RLT
↳ 2.0 V FV

4. ABSOLUTE MAXIMUM RATINGS

Power Dissipation	PD (Ta=25°C)	54mW
Forward Current	IF (Ta=25°C)	20mA
Peak Forward Current	IFP (Ta=25°C)	100mA
Reverse Voltage	VR (Ta=25°C)	5V
Operation Temperature	Topr	-40 ~ +85°C
Storage Temperature	Tstg	-40 ~ +100°C

*1 Duty 1/10, 1kHz

ZVD ↳ I=7mA
NO
R
$I_R = \frac{(5.0-2)V}{R} = .007A$
$R = 142.9\Omega \rightarrow 150\Omega$

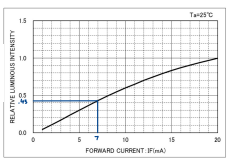
$P_{LED} = IV$
$= (.007A)(2.0\ V)$
$= .014\ W$

$P_{min} = I^2R$
$= (.007A)^2(150\ \Omega)$
$= .00735\ W$

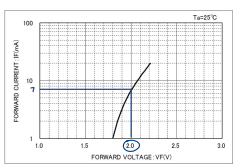
Resistance	150 Ohms
Tolerance	±1%
Power (Watts)	0.125W, 1/8W

∴ .1W, 1/10W

RELATIVE LUMINOUS INTENSITY - FORWARD CURRENT



FORWARD CURRENT - FORWARD VOLTAGE



Filtering Capacitor

$Z = \frac{1}{2\pi f_c}$

$f_1 = 1\ MHz$

$Z_1 = \frac{1}{2\pi(1e6\ Hz)(1e-7\ F)} = 1.59\ \Omega$

↳ lower impedance = better flow to GND

$f_2 = 10\ MHz$

$Z_2 = \frac{1}{2\pi(10^7\ Hz)(1e-7\ F)} = 159\ \Omega$

Bulk Capacitor

Load current = 50% Abs. Max current
= 50 mA

$I_{bulk} = 20\ mA$

Electrical characteristics

Nominal capacity (10h Rate, 2.5V cut-off)	245 mAh
Open circuit voltage (at + 20 °C)	3.0 V
Standard Continuous Discharge Current	0.5 mA
Maximum Continuous Discharge Current	5 mA
Maximum Pulse Discharge Current at 1 sec	20 mA
Nominal Energy	735 mWh
AC Impedance @ 1kHz	2.0 Ohm

$I_{load} = I_{batt} + I_{cap}$

$I_{cap} = I_{load} - I_{batt}$
$= 50\ mA - 20\ mA$
$= 10\ mA$

$C = \frac{Q}{V}$

$\Delta V = \frac{Q}{C} = \frac{It}{C}$ | $C = 47\ \mu F$

$= \frac{(.01\ A)(.002\ s)}{47e\text{-}6\ F}$

$= .425$

$V = 3.0 - .425 = 2.575\ V > 2.4\ V$

$V = 3.0 + .425 = 3.425\ V < 3.6\ V$

Table 2. STM32F030x4/x6/x8/xC family device features and peripheral counts

Peripheral	STM32 F030F4	STM32 F030G4	STM32 F030K6	STM32 F030C6	STM32 F030R8	STM32 F030C8	STM32 F030R8
Flash (Kbytes)	16	16	32	32	64	256	64	256
SRAM (Kbytes)	4			8	32	8	32
Timers	Advanced control	1 (16-bit)					
	General purpose	4 (16-bit)(1)			5 (16-bit)		
	Basic	-			1 (16-bit)(2)	2 (16-bit)	1 (16-bit)(2)	2 (16-bit)
	SysTick	1					
	Watchdog	2					
Comm. interfaces	SPI	1			2		
	I²C	1			2		
	USART	1			2(3)	6	2(3)	6
12-bit ADC (number of channels)	1 (9 ext. + 3 int.)	1 (10 ext. + 3 int.)	1 (8 ext. + 3 int.)	1 (10 ext. + 3 int.)	1 (16 ext. + 3 int.)	1 (19 ext. + 3 int.)	1 (16 ext. + 3 int.)	1 (19 ext. + 3 int.)
GPIOs	14	30	28	39	55	37	55	81
Max. CPU frequency	48 MHz							
Operating voltage	2.4 to 3.6 V							
Operating temperature	Ambient operating temperature: -40°C to 85°C / Junction temperature: -40°C to 105°C							
Packages	TSSOP20	UQFP32	LQFP48			LQFP64		

Table 33. HSE oscillator characteristics

Symbol	Parameter	Conditions(1)	Min(2)	Typ	Max(2)	Unit
I_{DD}	HSE current consumption	During startup(3)	-	-	8.5	mA
		V_{DD} = 3.3 V, Rm = 45 Ω, CL = 10 pF @8 MHz	-	0.9	-	
		V_{DD} = 3.3 V, Rm = 30 Ω, CL = 20 pF @32 MHz	-	1.9	-	
g_m	Oscillator transconductance	Startup	10	-	-	mA/V
$t_{SU(HSE)}$(4)	Startup time	V_{DD} is stabilized	-	2	-	ms